RADAR: Generation of a Displacement Map

"Jakarta – the sinking city?"

Introduction:

Jakarta is the capital city of Indonesia located on the cost of Java, at the mouth of the Ciliwung River[(1)]. The population of the city, around 11.6 million people living on 660 km^2 of land[(2)], has been rapidly growing since the 1950[(1)]. The city is facing many problems, like flooding and infrastructure damage, caused by the rapid rates of subsidence reported by many sources (Abidin et al., 2001; Situmorang et al. 2021; Whincup et al., 2023).

The subsidence of the city is said to be caused by multiple factors (Abidin et al., 2001). Firstly, the alluvial plain, that the city is based on, makes it quite susceptible to the problem, as the sediments underneath the city are prone to compacting under the weight of the developing infrastructure (Abidin et al., 2001). Secondly, unsustainable groundwater usage, caused by over-pumping in the city area and not giving the resources the possibility to recover, is causing the city to sink additionally (Whincup et al., 2023). Lastly, the general problem of climate change and rising sea levels is simply adding another layer to the problem.

SAR (Synthetic Aperture Radar) technology, with its capabilities to capture images of big areas during the day and night and almost regardless of the weather conditions, is commonly used for the purposes of disaster management (e.g.: floods, earthquakes), land monitoring (e.g.: military activities, land use) and monitoring ground deformation caused by natural or anthropogenic activities (Kim et al., 2025; Qu et al., 2015), which is the topic of this paper.

A SAR dataset contains of pixels where each of them is associated with a complex value which using signal processing is translated into the amplitude ('the intensity') and the phase ('the return angle') of a signal (Kim et al., 2025; Rosen et al., 2000) crucial for the ground displacement analysis. Lately, the problem of subsidence is approached with a time series analysis which offers a more reliable result (Qu et al., 2015), but before that a simpler approach using only two SAR images combined into a change interferogram was used for that purpose (Amelung et al., 1999). The latter approach will be used in this paper, even though spatial decorrelation of SAR acquisitions can influence the results quite severely (Qu et al., 2015) when we take into consideration the time difference between the two images being taken.

The goal of this paper is to research the use of SAR imagery to check the subsidence of the city of Jakarta in 2024 using a displacement map made based on two SAR images from the beginning and the end of the year.

Motivation for the choice of the area of interest:

I first heard about Jakartas situation in the last year of my bachelors, in the costal management class, and the story really caught my attention. Back then I really couldn't understand why all of the proposed solutions to the problems that the city is facing weren't actually dealing with the real causes of the problems, like the overuse of the groundwater in the area of the city. I found this case very interesting so this story really sticked with me for the past years.

Methods:

The analysis was completed in ESA SNAP using two Sentinel-1 level 1 SCL datasets from the 5th of January 2024 and the 30th of December 2024 based on the steps provided in the document "Radio Detection and Ranging -Practical- Generation of a Displacement Map" (*Hadj-Rabah, 2025)*. The names of files downloaded from the Copernicus Data Space are listed below:

- S1A_IW_SLC_1SDV_20240105T111536_20240105T111604_051970_0647AB_7D41
- S1A_IW_SLC_1SDV_20241230T111531_20241230T111559_057220_0709B7_B382.

In the first step of the analysis the files were loaded into ESA SNAP and the area of interest (sub-swatch (IW2) and bursts (1-3)) was chosen (Fig.1) and the orbit file was applied to correct any positioning errors of the sensor.

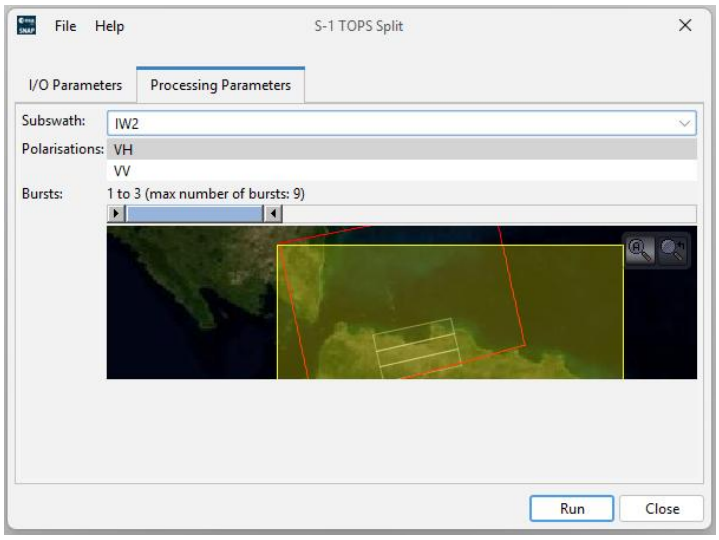

Figure 1: S-1 TOPS Split view from ESA SNAP when choosing the exact area of interest.

Second step included creating a single image (co-registration and back-geocoding) from the two datasets used as data input. This step is needed to check every pixel for its coordinates to make the two images perfectly (or almost perfectly) aligned. After that, an interferogram was created from the stacked pictures (Fig. 2).

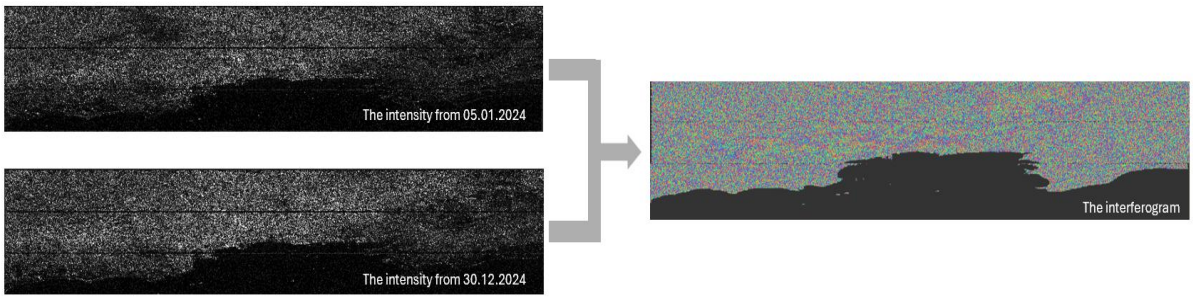

Figure 2: Process of creating of the interferogram from the stacked images.

Next step is connected to de-bursting the image, this step is needed to get rid of the boundaries between separate bursts of an image. After that the interferogram has to be

subjected to topographic phase removal, which is essential when working in mountainous areas as it removes the distortions that are caused by terrain (Fig. 3).

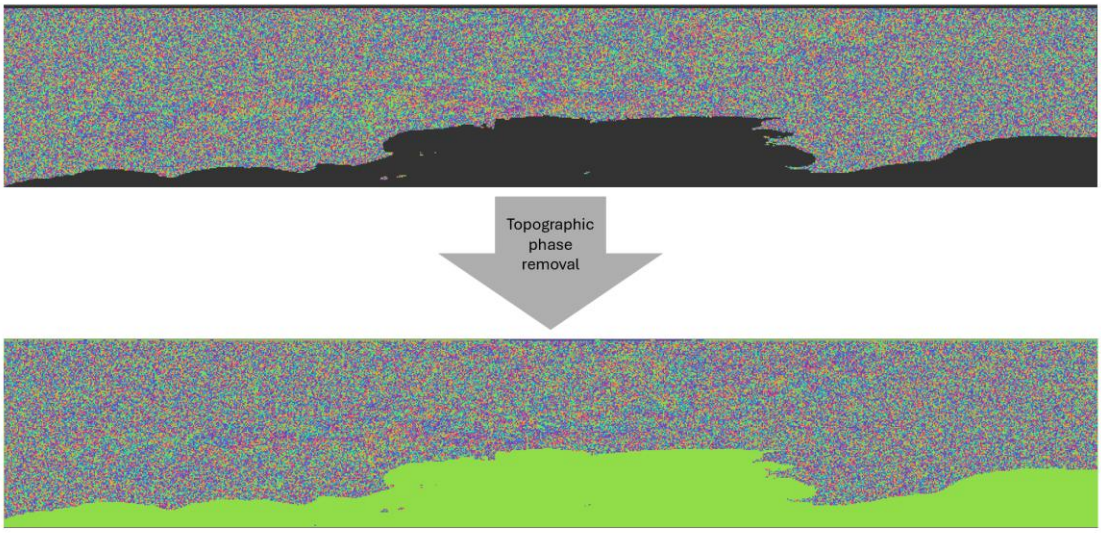

Figure 3: Showing interferogram before (top image) and after (bottom image) the topographic phase removal.

Multilooking is used in the next step to create an image with a square pixel size. In the parameters we need to specify how many looks in the range direction we want to assign to a given number of looks in the azimuth direction. This can lead to reduction of spatial resolution, while at the same time, leads to speckle reduction in the image[3].

Next step is the Goldstein Phase Removal which is supposed to enhance the phase unwrapping accuracy and remove some of the noise from the image. The result of multilooking as well as the applied filter is shown on Figure 4.

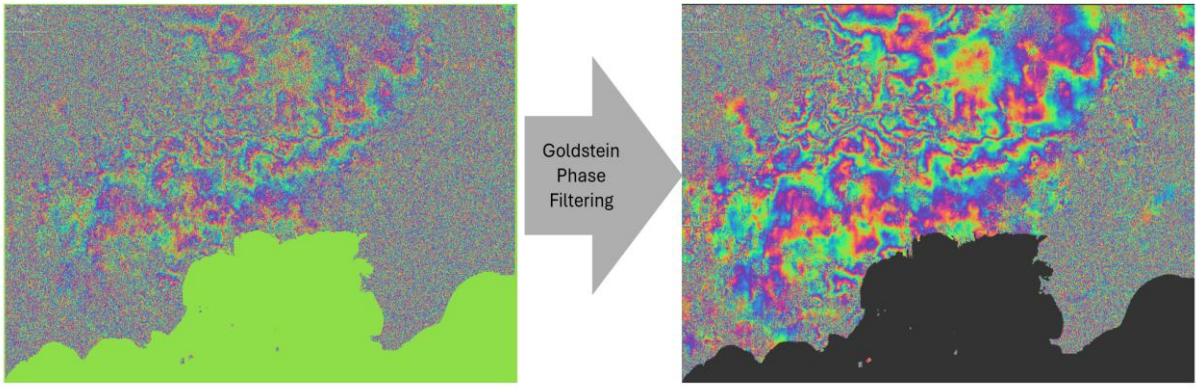

Figure 4: The image created after multilooking (number of range looks = 8, number of azimuth looks = 2) (left) and the image after applying the Goldstein Phase Filtering (coherence threshold = 0.8) (right).

Next steps were connected to SNAPHU which is an implementation of the Statistical-cost, Network-flow Algorithm for Phase Unwrapping proposed by Chen and Zebker[4]. SNAPHU algorithms goal is to compute the most likely unwrapped

solution for each pixel given the observable input data[4]. Firstly, SNAPHU had to be exported and run in the command line and then imported back to SNAP with the new unwrapped phase, that was created. Result of running SNAPHU is shown in Figure 5 (parameters changed: number of tile rows: 1, number of tile columns: 1).

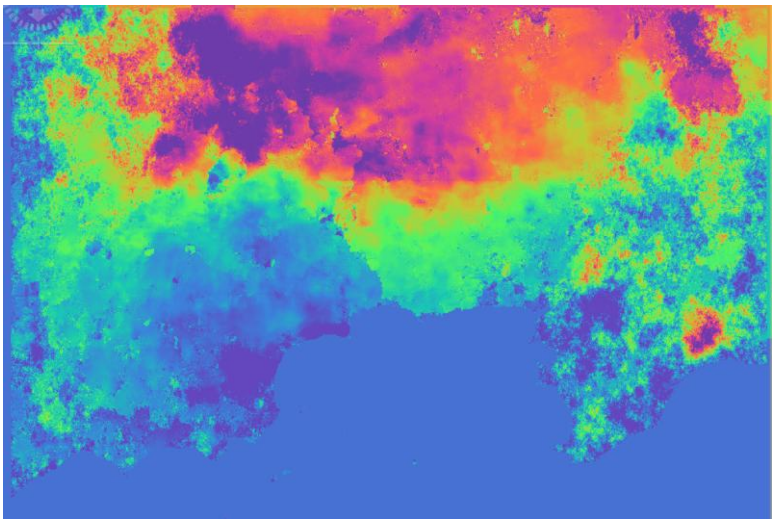

Figure 5: Image after the phase unwrapping.

In the next step the displacement is calculated based on the phase difference between the two points in time. The calculation is based on the equation[5]:

$$d = -\frac{\lambda}{4\pi}\Delta\phi_d$$

Where:

- **d** – is the relative scatterer displacement in the slant range direction,
- **λ** – is the transmitted wavelength,
- **$\Delta\phi_d$** – is the interferometric phase difference**.**

The last step is connected to georeferencing the displacement map and getting rid of most of the noise caused by topography of the area using the Range Doppler Terrain Correction (Fig. 6).

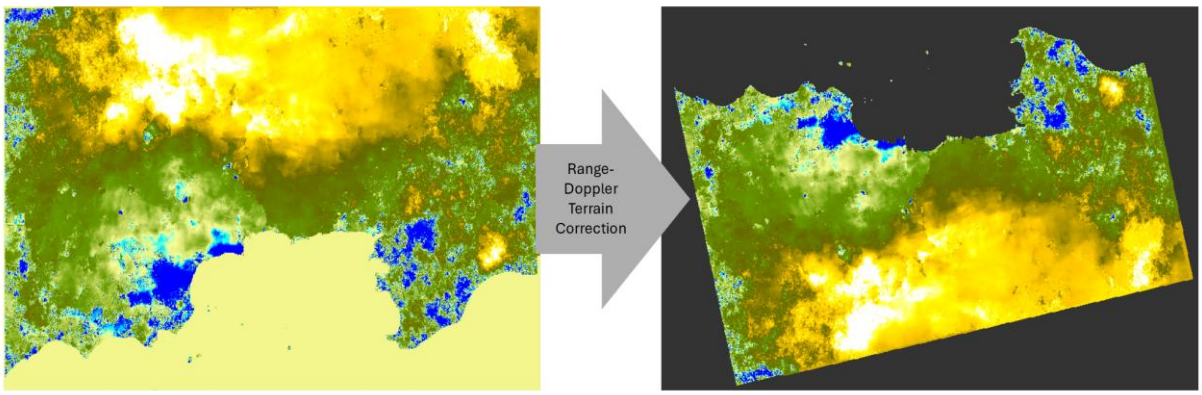

Figure 6: Showing the displacement map before (left) and after (right) the Range-Doppler Terrain Correction.

Results:

As a result of the analysis a displacement map of the region was obtained as well as a histogram of the resulting displacement values (Fig. 7). The values of displacement were between -7.4 cm and +20.8 cm, but the majority of the results (95%) was in between – 2.2 cm and +14.4 cm.

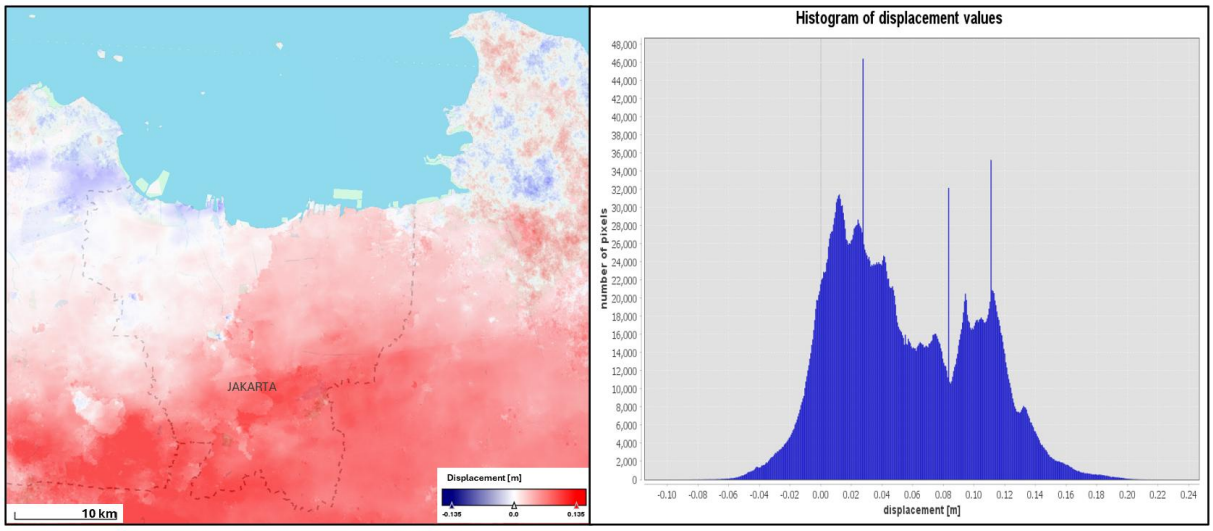

Figure 7: A map of Jakarta overlaid with the displacement map (left) and the histogram showing the distribution of results (right).

The biggest subsidence can be observed near the shore. Positive values of displacement can be seen in the southern part of the city. Most of the terrain has values of displacement higher than zero which means that the ground is rising.

Discussion:

The method of combining two SAR images, from the beginning and the end of 2024, into a differential interferogram resulted in a displacement map with the values that compering to the literature are quite high. Based on this map (Fig. 7) one could come to a conclusion that in 2024 most of Jakarta was actually rising and not sinking during that year. This result may be not completely truthful, as the trend in the years before 2024 is generally showing that the city is sinking rather than the opposite (Abidin et al., 2001; Situmorang et al., 2021).

The results indicate that the land is sinking the most near the shores, which from the geological and hydrographical standpoint makes sense, as the sediment from the river was the loosest further in the plain and so more susceptible to compaction under the heavy infrastructure as well as any kind of erosion by the ocean waters washing ashore.

In the paper by Situmorang et al. (2021), where the method used is similar to the one used in this paper, the results (Fig. 8) show similar trends as the results obtained here. We can see that the western part of the shore is sinking the fastest, with the entire west Jakarta characterized by the smallest values of displacement. In other parts of the city we can't observe that dramatic of a result as the southern part seems to be, similarly to results presented in this paper, rising. The results presented by Situmorang et al. (2021) are not very

easily interpretable as the color pallet used for showing the displacement is not centered around zero.

The real difference between the two results (Fig. 8) though are the values of displacement as in this work most of them falls between -2.2 cm and +14.4 cm which are very different to the values obtained by the other authors. In their case most of the area is clearly sinking and only a small part is rising slightly with the maximum values of +1.8 cm.

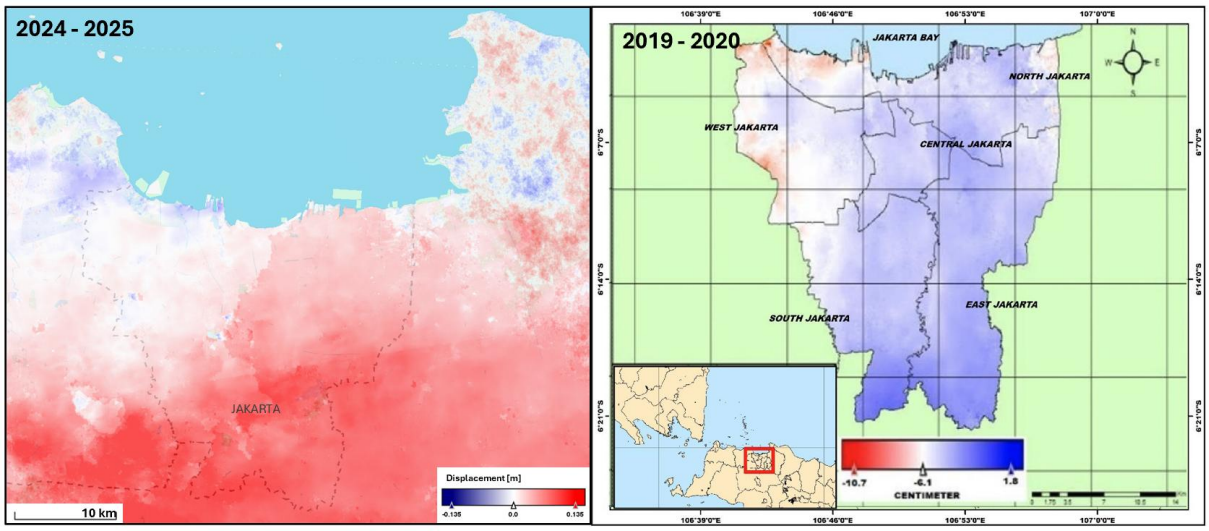

Figure 8: Showing the results of this paper (left side) and the results from the paper "'Land subsidence detection in Jakarta Province using sentinel-1a satellite imagery" by Situmorang et al. (2021) (right side).

The difference in the values may be connected to slightly different algorithms used during the process or maybe from a mistake during the processing of the image (as I'm only doing in for the first time), but also on a more positive note, the results obtained in this paper could mean that the sinking of Jakarta has drastically stopped in the time between 2020 and 2024, which is not a very probable theory but quite an optimistic one.

Sources:

List of internet sources:

(1) *Jakarta Population 2025*
(2) *Jakarta - Wikipedia*
(3) *https://step.esa.int/docs/tutorials/S1TBX%20SAR%20Basics%20Tutorial.pdf*
(4) *https://step.esa.int/main/snap-supported-plugins/snaphu/*
(5) *https://step.esa.int/main/wp-content/help/versions/11.0.0/snap-toolboxes/eu.esa.microwavetbx.sar.op.insar.ui/operators/PhaseToDisplacementOp.html*

List of literature:

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Amelung, F. et al. (1999) 'Sensing the ups and downs of Las Vegas: Insar reveals structural control of land subsidence and aquifer-system deformation', Geology, 27(6), p. 483. doi:10.1130/0091-7613(1999)027<0483:stuado>2.3.co;2.

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Situmorang, D., Arhatin, R.E. and Lumban-Gaol, J. (2021) 'Land subsidence detection in Jakarta Province using sentinel-1a satellite imagery', IOP Conference Series: Earth and Environmental Science, 944(1), p. 012036. doi:10.1088/1755-1315/944/1/012036.

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